

Final Report Guidance (ELEC340/440)

Your final report should clearly demonstrate **your contribution to the project**.

It must summarise:

- Your **results and key achievements**
- How you met your **original objectives**
- How your work relates to the **state of the art**
- Your **recommendations for future work**
- Your **professional development and reflection**

This report should resemble an **industry-standard technical report**, not an academic dissertation.

Technical reports are:

- Concise and focused
- Results-driven
- Structured for professional readers
- Clear and evidence-based
- Forward-looking (with recommendations)

The **main sections of your report must be less than 4000 words in total** (excluding appendices, acknowledgements, and references).

Professional Report Writing Standards

As future engineers, your report should align with professional expectations such as those promoted by the Institution of Engineering and Technology (IET), the professional organisation for Electronic Engineers.

The IET suggests “10 Laws of Good Report Writing”¹

:

1. Produce the report for your reader(s)
2. Keep the report as short as possible
3. Organise information for the convenience of the reader

4. Include accurate references
5. Ensure your writing is accurate, concise and straightforward
6. Include diagrams with the right labels in the right place for your reader
7. Make sure your summary gives the whole picture in brief
8. Check the report for technical errors, typing errors and inconsistencies
9. Consider design as well as content
10. Produce the report for your reader(s)

(Points 1 and 10 are intentionally identical to emphasise their importance)

The focus of your report is on demonstrating your contribution to the project.

The Structure of the Report

The structure of your report should reflect the nature of your project. The sections below are typical for a final-year report, but you should discuss any variations with your supervisor.

If you need additional support, consult the University's [Student Knowhow](#) pages for academic writing guidance as well as one-to-one appointments to discuss your work.

Title page

Use the template provided on Canvas

Acknowledgements

Not compulsory and not included in the word count

Summary (Executive Summary)

- Maximum of 200 words
- Text only (no figures)
- Focus on key achievements and outcomes
- Should provide a complete overview of the project

(You have already prepared this for Bench Inspection)

Tables of Contents, Figures, Tables, Abbreviations, etc.

These need to be included where appropriate

Introduction /Scope

- Set the context for your project

- Define your key objectives
- Explain how your work compares to the field

(While a full literature review is NOT required, this section should reference the state of the art.)

Procedure / Methodology

Provide a clear summary of your technical approach

- Experimental set-up or system design
- Implementation strategy
- Evaluation methods
- How progress was measured

The description should be detailed enough to allow someone to reproduce your work. However, the long equipment list or component specifications should be placed in the appendix.

Findings / Results

Present your key results clearly:

- Data, performance metrics, validation
- Comparison with previous work
- Evidence of your contribution

Focus on what you achieved and how it advances or improves existing solutions.

Conclusions

Analyse your overall progress:

- Have you delivered novel results?
- Have you improved on previous work?
- How do you differentiate from competitors?
- To what extent did you meet your objectives?

This section should put your work into its broader technical context.

Recommendations / Future work

Technical reports should always look forward. Consider,

- How could the project be developed further?

- What technical improvements are possible?
- What would be the next engineering steps?

Reflection

Reflect on your professional development:

- What new technical and professional skills did you develop?
- How does this compare with your original skills audit?
- How has this project prepared you for industry or further study?

References / Bibliography

You must reference all external sources. Failure to attribute sources correctly may constitute **academic misconduct**.

- Any word-for-word quotation must appear in quotation marks.
- All sources must appear in the reference list.

The **IEEE referencing format is strongly recommended**. Common formats include:

Books:

Author, "Title of chapter in the book," in Title of Published Book, xth ed. City of Publisher, (only U.S. State), Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx-xxx

Conference articles

J. K. Author, "Title of chapter in the book," in Title of Published Book, xth ed. City of Publisher, (only U.S. State), Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx-xxx

Online materials (including datasheets)

J. K. Author or Name of Company., City of Co. Abbrev. State, Country). Title of datasheet/Manual/Handbook/webpage. Accessed: Date. [Online]. Available: <http://www.url.com>

Periodicals and academic Journals

J. K. Author, "Title of paper," Abbrev. Name of Periodical, vol. x, issue. x, pp. xxx-xxx, Abbrev. Month, year, doi: xxx.

Software and Code

For many projects, software or firmware is a key contribution; however, source code is rarely included in a technical report since it can be difficult to follow and requires in-depth knowledge of the programming language that is used. Flow charts, block diagrams or pseudocode can be a much better way of explaining how your software works, while the full source code can be included in an appendix.

Many software projects make use of third-party libraries and toolboxes. It is crucial that you correctly attribute this, since omitting references can be academic misconduct. Clearly indicate what parts you developed yourself, what parts are adapted from third-party sources and where you are just using a third-party source (e.g. a library).

The Force11 software citation working group has developed principles for citing software sources². Please follow their guidance.

Data sources

If you use third-party data sources (e.g. image databases or any other datasets that you have not recorded yourself), please make sure you attribute them correctly.

The Force11 has developed guidance on how to cite data sources³.

Appendices

Appendices are a good way to include more detailed information that is not relevant to all readers of the report. Since they are not part of the main report, they are not included in the word limit.

For example, your supervisor and assessor may not be interested in reading 10 pages of source code, but a student trying to reproduce your findings will appreciate this.

Similarly, large datasets, detailed circuit diagrams, full bill of materials, lists of equipment, etc, are best included as appendices and referenced in the main text.

There could be a wide range of appendices giving details on things such as:

- Source code,
- Full circuit diagrams
- Bill of materials
- Recorded datasets / raw data
- Instruction manuals
- Technical drawings
- Anything else you might want to include

Illustrations, Figures and Tables

Figures can significantly improve clarity.

“A picture is worth a thousand words”⁴

Figures, graphs, tables or equations can be placed throughout the report, but there are a few things you want to keep in mind.

Illustrations can often summarise information much better than text, so consider if this can improve your report as well. However, consider how you can make your figure the most relevant. E.g. is a photo of a multimeter showing a reading the most relevant way of showing this, or would a table of readings be more informative?

Do you need a full circuit diagram or would a section showing only the critical part you discuss be more appropriate (while the full diagram can be added as an appendix).

For graphs, please make sure all your axes are labelled correctly in SI units.

Please make sure your figures, graphs, and tables are well designed, informative and related to the main text of the report. All illustrations should be referenced in the text and listed in the relevant table. If you are using third-party figures, please make sure they are referenced correctly.

References

¹ J. van Emden, J. Eastel revised by A. Kerr “A Guide to Technical Report Writing” [online] <https://www.theiet.org/media/5182/technical-report-writing.pdf> [Accessed online 29th August 2025]

² Smith, Arfon M., Daniel S. Katz, and Kyle E. Niemeyer. "Software citation principles." *PeerJ Computer Science* 2 (2016): e86. DOI: [10.7717/peerj-cs.86](https://doi.org/10.7717/peerj-cs.86)

³ M. Artone. “Joint declaration of data citation principles–FINAL”. FORCE11. 2014 Oct 21. <https://doi.org/10.25490/a97f-egy4>

⁴ A. Brisbane, "Newspaper Copy That People Must Read, Advertising's Relation to the Growth of Reading Ability—the Thunderstorm and "Yellow" Journalism—an Example of the Power of Comparison in Writing", *Printers' Ink, A Journal for Advertisers*. April 20, 1911. p. 17.